

Reliability From α to ω : A Tutorial

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Reliability is a fundamental problem for measurement in all of science. Although defined in multiple ways, and estimated in even more ways, the basic concepts seem straightforward and need to be understood by practitioners as well as methodologists. Reliability theory is not just for the psychometrician estimating latent variables, it is for everyone who wants to make inferences from measures of individuals or of groups. For the case of a single test administration, we consider multiple measures of reliability, ranging from the worst (β) to average (α , λ_3) to best (λ_4) split half reliabilities, and consider why model-based estimates (ω_h , ω_r) should be reported. We also address the utility of test–retest and alternate form reliabilities. The advantages of immediate versus delayed retests to decompose observed score variance into specific, state, and trait scores are discussed. But reliability is not just for test scores, it is also important when evaluating the use of ratings. Estimates that may be applied to continuous data include a set of intraclass correlations while discrete categorical data needs to take advantage of the family of κ statistics. Examples of these various reliability estimates are given using state and trait measures of anxiety given with different delays and under different conditions. An [online supplemental materials](#) is provided with more detail and elaboration. The [online supplemental materials](#) is also used to demonstrate applications of open source software to examples of real data, and comparisons are made between the many types of reliability.

Public Significance Statement

A tutorial on the estimation of the reliability of test scores considers classical and model based approaches. Examples using open source software applied to several real world data sets are provided.

Keywords: reliability, generalizability, classical test theory, R packages

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Reliability is a fundamental problem for measurement in all of science for “(a)ll measurement is befuddled by error” (McNemar, 1946, p. 294). Perhaps because psychological measures are more befuddled than those of the other natural sciences, psychologists have long studied the problem of reliability (Cronbach, 1951; Feldt & Brennan, 1989; Guttman, 1945; Kuder & Richardson, 1937; Lord, 1955; McDonald, 1999; Spearman, 1904b), and it remains an active topic of research (Bentler, 2009; McNeish, 2018; Revelle & Zinbarg, 2009; Sijtsma, 2009; Wood, Harms, Lowman, & DeSimone, 2017). Unfortunately, although recent advances in the

theory and measurement of reliability have gone far beyond the earlier contributions, much of this literature is more technical than readable and is aimed for the specialist rather than the practitioner. (This is not a new problem; e.g., Anastasi, 1967; Glass, 1986, bemoan this tendency). We hope to remedy this issue somewhat, for an appreciation of the problems and importance of reliability is critical to the activity of measurement across many disciplines. Reliability theory is not just for the psychometrician estimating latent variables, but also for the baseball manager trying to predict how well a high performing player will perform the next year, for accurately estimating agreement among doctors in patient diagnoses, and in evaluations of the extent to which stock market advisors underperform the market.

Issues of reliability are fundamental to understanding how correlations between observed variables are (attenuated) underestimates of the relationships between the underlying constructs, how observed estimates of a person’s score are biased estimates of their latent score, and how to estimate the confidence intervals around any particular measurement. Understanding the many ways to estimate reliability as well as the ways to use these estimates allows one to better assess individuals and to evaluate selection and prediction techniques. This is not just a problem for measure-

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ment specialists but for all who want to make theoretical inferences from observed data. Schmidt and Hunter (1996) discuss 26 ways that not correcting for the effects of reliability and measurement error can hinder progress in many areas of psychological research. However, Borsboom and Mellenbergh (2002) take a contrary position and suggest that using classical measures of reliability for such corrections is an error.

The fundamental question in reliability is to what extent do scores measured at one time and place with one instrument predict scores at another time and/or place and perhaps measured with a different instrument? That is, given a person's score on Test 1 at Time 1, what score should be expected at a second measurement occasion? The naive belief is that if the tests measure the same construct, then people will do just as well on the second measure as they did on the first. This mistaken belief contributes to several errors including the common view that punishment improves and rewards diminish subsequent performance (Kahneman & Tversky, 1973) and other popular phenomena like the "sophomore slump" and the "Sports Illustrated jinx" (Schall & Smith, 2000). More formally, the expectation for the second measure is just the regression of observations at Time 2 on the observations at Time 1. If both the Time 1 and Time 2 measures are equally "befuddled by error," then the observed relationship is the reliability of the measure: the ratio of the latent score variance to the observed score variance.

Reliability as a Variance Ratio

The basic concept of reliability seems to be very simple: observed scores reflect an unknown mixture of signal and noise. To detect the signal, we need to reduce the noise. Reliability thus defined is a function of the ratio of signal to noise. The signal might be something as esoteric as a gravity wave produced by a collision of two black holes, or as prosaic as estimating the expected batting average of a baseball player based upon the performance of the prior year. The noise in gravity wave detectors include the seismic effects of cows wandering in fields near the detector as well as passing ice cream trucks. The noise in batting averages include the effect of opposing pitchers, variations in wind direction, and the effects of jet lag and sleep deprivation. We can enhance the signal/noise ratio by either increasing the signal or reducing the noise. Unfortunately, this classic statement of reliability ignores the need for unidimensionality of our measures and equates expected scores with construct scores, a relationship that needs to be tested rather than assumed (Borsboom & Mellenbergh, 2002).

We can credit Spearman (1904b) for the original formalization of reliability. In the first of two landmark papers (the other, Spearman, 1904a, laid the basis for factor analysis and measurement of cognitive ability) he developed the ordinal correlation coefficient and the basic principles of reliability theory. Spearman's fundamental insight was that an observed test score could be decomposed into two unobservable constructs: a *latent* score of interest and a residual but latent *error* score:

$$X = \chi + \epsilon. \quad (1)$$

Reliability was defined as the fraction of an observed score variance that was not error:

$$r_{xx} = \frac{V_X - \sigma_\epsilon^2}{V_X} = 1 - \frac{\sigma_\epsilon^2}{V_X}. \quad (2)$$

If the test is unidimensional the product of the observed score variance and the reliability is an estimate of the variance of the *latent construct* (sometimes called the "true score")¹ in a test that we will symbolize² as $\sigma_\chi^2 = V_X - \sigma_\epsilon^2 = r_{xx}V_X$. Spearman (1904b) developed reliability theory because he was interested in correcting the observed correlation between two ability tests for their lack of reliability. In modern terminology, this disattenuated correlation ($\rho_{\chi\eta}$) represents the correlation between two latent variables (χ and η) estimated by the correlation of two observed tests (r_{xy}) corrected for the reliability of the observed tests (r_{xx} and r_{yy} ; see Figure 1).

$$\rho_{\chi\eta} = \frac{r_{xy}}{\sqrt{r_{xx}r_{yy}}}. \quad (3)$$

Furthermore, given an observed score, the variance of the error of that score (σ_ϵ^2) is just the observed test variance times one minus the reliability and thus the standard deviation of the error associated with that score (the *standard error of measurement*) is:

$$\sigma_\epsilon = \sigma_x \sqrt{1 - r_{xx}}. \quad (4)$$

Although expressed as a correlation between observed scores, reliability is a ratio of reliable variance to total variance. In addition, because the covariance of the latent score with the observed score is just the reliable variance, the predicted latent score $\hat{\chi}$ is

$$\hat{\chi} = r_{xx}x \quad (5)$$

where x is the raw deviation score ($x = X - \bar{X}$). From Equation 4, we know the standard error of measurement and can give a confidence interval for our observed score and even for the estimated latent score (Charter & Feldt, 2001):

$$\hat{\chi}_i = r_{xx}x_i \pm t_{\alpha/2, df} \sigma_x \sqrt{r_{xx}(1 - r_{xx})} \quad (6)$$

where $t_{\alpha/2, df}$ represents Student's t with an appropriate probability level (e.g., $\alpha = .05$).

Increasing reliability reduces the standard error of measurement (Equation 4) and increases the observed correlation with external variables (Equation 3). That is, if we knew the reliabilities, we could correct the observed correlation to find the latent correlation and estimate the precision of our measurement. The problem for Spearman was, and remains for us today, how to find reliability?

Equations 1–6 are intellectually interesting, but not very helpful, for they decompose an observed measure into the two unobservable variables of latent score and latent error. To make it even more complicated, all tests are assumed to measure something stable over time (denoted as T for *trait* like), something that varies over time (reflecting the current *state* and denoted as S), some specific variance (s) that is stable but does not measure our trait of interest, and some residual, random error (E ; Baltes, 1987; Cattell,

¹ As discussed by Lord and Novick (1968) and Borsboom and Mellenbergh (2002), classical test theories "true score" is just the expected value and should not be confused with Platonic Truth.

² Observed variables and correlations are shown in conventional Roman fonts, latent variables and latent paths in Greek fonts.

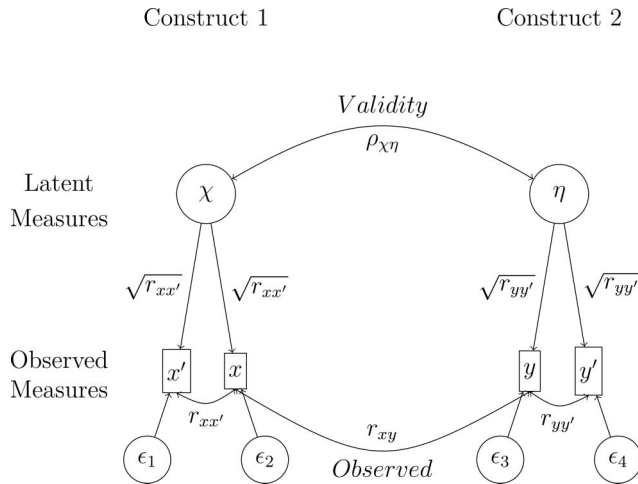


Figure 1. The basic concept of reliability and correcting for attenuation. Adjusting observed correlations (r_{xy}) by reliabilities ($r_{xx'}$, $r_{yy'}$) estimates underlying latent correlations ($\rho_{\chi\eta}$; see Equation 3). Observed variables and correlations are shown in conventional Roman fonts, latent variables and latent paths in Greek fonts.

1966b; Hamaker, Schuurman, & Zijlman, 2017; Kenny & Zautra, 1995).

Although ultimately interested in the precision of a score for each individual, reliability is expressed as a ratio of variances between individuals:³ The reliability of X (r_{xx}) is just the percentage of total variability (V_X) that is not error. Unless we have repeated measures, error is an unknown mixture of variability due to differences in item means, the Person $\times$ Item interaction, and some unmodeled residual. The reliable between person variance is a mixture of Trait, State, and specific variance. (For instance, an item such as “I enjoy a lively party” is an unknown mixture of trait extraversion, state positive affect, and the specific wording of the item—how one interprets lively and party.) If we are interested in the stable trait scores, reliability is the ratio of (unobservable) trait variance (σ_T^2) to (observable) total test variance (V_X). (We use σ^2 to represent unobservable variances, V to represent observable variance.) That is,

$$r_{tt} = \frac{\sigma_T^2}{V_X} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_S^2 + \sigma_s^2 + \sigma_e^2}. \quad (7)$$

However, if we are interested in how well we are measuring a particular fluctuating state (e.g., an emotion) we want to know

$$r_{ss} = \frac{\sigma_S^2}{V_X} = \frac{\sigma_S^2}{\sigma_T^2 + \sigma_S^2 + \sigma_s^2 + \sigma_e^2}. \quad (8)$$

The problem becomes how to find σ_T^2 or σ_S^2 and how to separate their effects. Although trait scores are thought to be stable over time, state scores, while fluctuating, show some (unknown) short-term temporal stability. Consider a measure of depression. Part of an individual's depression score will reflect long-term trait neuroticism and some of it reflects current negative emotional state. Two measures taken a few hours apart should produce similar trait and state values, although measures taken a year apart should reflect just the trait.

In all cases, we are interested in the scores for the individuals being measured. To make the problem even more complicated, it is likely that our trait or state scores reflect some aggregation of item responses or of the ratings of judges. Thus, we want to assess the variance due to traits or states that is independent of the effects of items or judges, how much variance is due to the items or judges, and finally how much variance is due to the interactions of items/judges with the trait/state measures.⁴ To be consistent with much of the literature, we will treat trait and state as both latent sources of variance for the observed score X and refer to trait as a stable across time and state as varying across time. We recognize, of course that traits do change over longer periods of time but will use this stable/unstable distinction for relatively short temporal durations. Although some prefer to think of specific variance (σ_s^2) and error variance (σ_e^2) as hopelessly confounded, we prefer to separate them for there are some designs (e.g., test–retest vs. parallel forms) that allow us to distinguish them.

Reliability as defined in Equations 7 and 8 is not just a function of the test, but also of who is being tested, where they are tested, and when they are tested. Because it is a variance ratio, increasing between-person variance without increasing the error variance will increase the reliability. Similarly, decreasing between-person variance will decrease reliability. Generalizability theory (Cronbach, Rajaratnam, & Gleser, 1963; Gleser, Cronbach, & Rajaratnam, 1965) is one way to estimate the individual variance components rather than their ratio. Another approach is Item Response Theory (e.g., Embretson, 1996; Lord & Novick, 1968; Lumsden, 1976; Mellenbergh, 1996; Rasch, 1966; Reise & Waller, 2009), which addresses this problem by attempting to get a measure of precision for a person's estimate that is independent of the variance of the population and depends upon just the probability of a particular person answering particular items.

Consistency, Reliability and the Data Box

When Cattell (1946) introduced the *data box* it was a three-way organization of measures taken over people, tests, and time. In typical Cattellian fashion, over the years this simple idea grew to as many as 10 dimensions (Cattell, 1966a; Cattell & Tsujioka, 1964). However, the three primary distinctions are still useful today (Nesselroade & Molenaar, 2016; Revelle & Wilt, 2016). Using these dimensions, Cattell (1964) distinguished between three ways that tests can be consistent: across occasions (reliability), across items (homogeneity), and across people (transferability or hardiness). We consider the first two of these concepts and leave the latter to a discussion of validity. These various types of reliability may be summarized graphically in terms of latent traits, paths, observed variables, and correlations (see Figure 2).

³ We can also find within-subject reliability across time. This will be discussed later and in the [online supplemental materials](#).

⁴ Unfortunately, some prefer to use State to reflect the measure at a particular time point and to decompose this “State” into Trait and Occasion components (Cole et al., 2005).

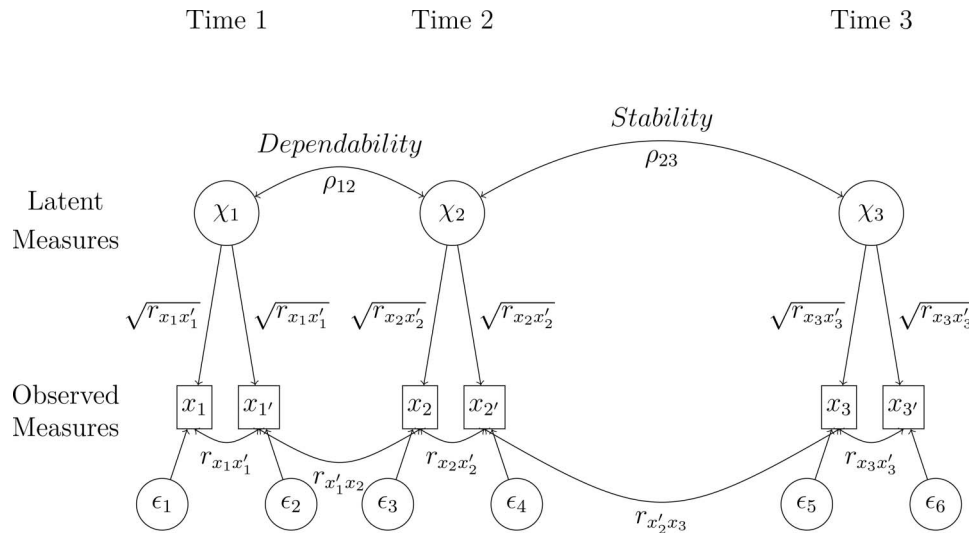


Figure 2. Reliability has many forms. Items within the same test provide estimates of *internal consistency*. Observed correlation between *alternate forms* or parallel tests given at the same time ($r_{x_1x_1'}$, $r_{x_2x_2'}$) estimate parallel test reliability. Tests given at (almost) the same time (Times 1 and 2; e.g., $r_{x_1x_2}$, $r_{x_1'x_2'}$) provide measures of *dependability*, while measures taken over a longer period (Times 2 and 3; e.g., $r_{x_1x_3}$, $r_{x_2x_3}$) are measures of *stability*. These measures differ in the amount of Trait and State and specific variance in the measures. Observed variables and correlations are shown in conventional Roman fonts, latent variables and latent paths in Greek fonts.

Alternative Estimates of Reliability

Test–Retest of Total Scores

Perhaps the most obvious measure of reliability is the correlation of test with the same test some time later. For Guttman (1945), this was reliability. If we have only two time points (t_1 and t_2), this correlation is an unknown mixture of trait, state, and specific variance and, additionally, a function of the length of time between the two measures:

$$r_{t_1t_2} = \frac{\sigma_T^2 + \tau^{t_2-t_1}\sigma_S^2 + \sigma_e^2}{\sigma_T^2 + \sigma_S^2 + \sigma_e^2} \quad (9)$$

where τ (the auto-correlation due to short term state consistency) is less than 1 and thus the state effect ($\tau^{t_2-t_1}\sigma_S^2$) will become smaller the greater the time lag. If the intervening time is long enough that the state effect is minimal, we will still have specific variance, and the correlation of a test with the same test later is

$$r_{xx} = \frac{\sigma_T^2 + \sigma_e^2}{\sigma_T^2 + \sigma_S^2 + \sigma_e^2} = \frac{\sigma_T^2 + \sigma_e^2}{\sigma_T^2 + \sigma_S^2 + \sigma_e^2}. \quad (10)$$

An impressive example of a correlation of the same measure over time is the correlation of .66 of ability as measured by the Moray House Exam at age 11 with the same test given to the same participants 69 years later when they were 80 years of age (Deary, Whiteman, Starr, Whalley, & Fox, 2004). This correlation was partially attenuated due to restriction of range for the 80-year-old participants. (The less able 11 year olds were less likely to appear in the 80-year-old sample.) When correcting for

this restriction (Sackett & Yang, 2000), the correlation was found to be .73.

But the Scottish Longitudinal Study is unusually long, and is it more common to take test–retests over much shorter periods of time. In most cases it is important that we do not assume that the state effect is 0 (Chmielewski & Watson, 2009). It is more typical to find a pattern of correlations diminishing as a function of the time lag but not asymptotically approaching zero (Cole, Martin, & Steiger, 2005; Damian, Spengler, Sutu, & Roberts, 2018). This pattern is taken to represent a mixture of stable trait variance and diminishing state effects such that the test–retest reliability across two time periods as shown in Equation 9 will become smaller the greater the time lag. Unfortunately, with only two time points we cannot distinguish between the trait and state effects. However, with three or more time points ($t_1, t_2, t_3, \dots, t_n$), we can decompose the resulting correlations ($r_{x_1x_2}, r_{x_1x_3}, r_{x_2x_3}, \dots$), into trait and state components using Structural Equation Modeling (SEM) procedures (Hamaker et al., 2017) or simple path tracing rules (Chmielewski & Watson, 2009) and the resolution continues to improve with four or more time points (Cole et al., 2005; Kenny & Zautra, 1995). (See Figure S1 in the online supplemental materials).

A large test–retest correlation over a long period of time indicates temporal *stability* (Boyle, Stankov, & Cattell, 1995; Cattell, 1964; Chmielewski & Watson, 2009). This should be expected if we are assessing something trait-like (such as cognitive ability or perhaps emotional stability or extraversion) but not if we are assessing something thought to represent an emotional state (e.g., alertness or arousal). Because we are talking about correlations, mean levels can increase or decrease

over time with no change in the correlation.⁵ Measures of trait stability are a mixture of immediate test–retest *dependability* and longer-term trait effects (Cattell, 1964; Chmielewski & Watson, 2009). For Boyle et al. (1995) and Cattell (1964), dependability was the immediate test–retest correlation; for Chmielewski and Watson (2009) the time lag of 2 weeks is considered an index of dependability. To Wood et al. (2017), dependability is assessed by repeating the same items later in one testing session.

All of these indicators of dependability and stability are in contradiction to the long-held belief that a problem with test–retest reliability is that it introduces memory effects of learning and practice (Kuder & Richardson, 1937). As evidence for the memory effect, Wood et al. (2017) reports that the average response times to the second administration of identical items in the same session is about 80% of the time of the first administration.

In the online supplemental materials (see Table S1) we compare multiple estimates of reliability for three different example data sets available in the *psychTools* package (Revelle, 2019b) in the R open-source statistical system (R Core Team, 2019). We describe these data sets in some detail as they are useful demonstrations of trait and state variations. We compare immediate retest to short (45 min) and then longer delay (1 to 7 days) on 10 mood items and 9–24 item trait scales for 1 to 4 weeks.

To compare the effects of an immediate retest versus a short delay versus a somewhat longer delay, consider the *msqR*, *sai*, and *tai* example data sets⁶; the analyses discussed below are demonstrated in the [online supplemental materials](#). We will use some of these items in the subsequent examples evaluating and comparing the immediate dependability and the 45 min and multiday stability coefficients of these measures. 10 items were given as part of the STAI and then immediately given again (with a slightly different wording) as part of the MSQ. Five of the items were scored in a positive (anxious) direction, five in the negative (nonanxious) direction.

Components of Variance Estimated by Test–Retest Measures

A powerful advantage of repeating items is that it allows for an assessment of subject consistency across time (the correlation for each subject of their pattern of responses across the two administrations) as well as the consistency of the items (the correlation across subjects of responses to each item; DeSimone, 2015; Wood et al., 2017). This allows for identification of unstable items and inconsistent responders. In addition, by using multilevel analyses⁷ it is possible to estimate the variance components due to people, items, the Person $\times$ Item interaction, time, the Person $\times$ Time interaction, and the residual (error) variance (DeSimone, 2015; Revelle & Wilt, 2019; Shrout & Lane, 2012). This is implemented for example as the Test–Retest function in the *psych* package. The responses to any particular item can be thought to represent multiple sources of variance, and the reliability of a test made up of items is thus a function of those individual sources of variance. If we let P_i represent the i_{th} person, I_j the j_{th} item, T_k the first or second administration of the item, then the response to any item (e.g., of an anxiety inventory) is a function of the mean level of all of the items in the test, the trait level of the person taking the item, the difficulty or average endorsement frequency of the item,

any temporal effects of a first or second administration, and all the possible interactions between these terms:

$$X_{ijk} = \mu + P_i + I_j + T_k + P_i I_j + P_i T_k + I_j T_k + P_i I_j T_k + \epsilon. \quad (11)$$

With complete data, we can find these components using conventional repeated measures analysis of variance of the data (i.e., *aov* in core R) or using multilevel functions such as *lmer* in the *lme4* package (Bates, Maechler, Bolker, & Walker, 2015) for R. As an example of such a variance decomposition consider the 10 overlapping mood items discussed in the online supplemental materials (see Table S1). Nineteen percent of the variance of the anxiety scores was due to between-person variability, 25% to the very short period of time, 19% to the interaction of person by time, and so forth; and 13% was residual (unexplained) variance.

Multilevel modeling approaches are particularly appropriate if repeating the same measure multiple times (e.g., in an experience sampling study; Bolger & Laurenceau, 2013; Fisher, 2015; Mehl & Conner, 2012; Mehl & Robbins, 2012; Wilt, Funkhouser, & Revelle, 2011; Wilt, Bleidorn, & Revelle, 2016, 2017). We can derive multiple measures of reliability, across subjects, across time, across items and the various Person $\times$ Time, Person $\times$ Items, Time $\times$ Item interactions (Cranford et al., 2006; Shrout & Lane, 2012). This is implemented in the *multilevel.reliability* function and discussed in more detail in a tutorial for analyzing dynamic data (Revelle & Wilt, 2019) as well as in the [online supplemental materials](#). Although these variance components can be found using traditional repeated measures analysis of variance, it is more appropriate to use multilevel techniques, particularly in the case of missing data.

Stability needs to be adjusted for dependability, and thus the .36 stability over 2 days of the SAI (online supplemental materials Table S1) should be adjusted for the immediate dependability of .85 to suggest a 2-day stability of anxious mood of .42, which is notably similar to that of the state–trait correlation of .43. When measuring mood, we need to disentangle the episodic memory components of the state measure from the semantic memory

⁵ For example, participants in the Scottish Longitudinal Study performed better in adulthood than they did as 11 year olds but the correlations showed remarkable stability.

⁶ The data ($N > 4,000$) were collected as part of a long-term series of studies of the interrelationships between the stable personality dimensions of extraversion, neuroticism, and impulsivity (e.g., Eysenck & Eysenck, 1964; Revelle, Humphreys, Simon, & Gilliland, 1980), situational stressors (caffeine, time of day, movie-induced affect, e.g., Anderson & Revelle, 1983, 1994; Rafaeli, Rogers, & Revelle, 2007; Rafaeli & Revelle, 2006), momentary affective and motivational state (e.g., energetic and tense arousal, Thayer, 1978, 1989; state anxiety, Spielberger, Gorsuch, & Lushene, 1970), and cognitive performance (Humphreys & Revelle, 1984). The Motivational State Questionnaire (MSQ, Revelle & Anderson, 1998) included 75 items taken from a number of mood questionnaires (e.g., Larsen & Diener, 1992; Thayer, 1978; Watson, Clark, & Tellegen, 1988) and had 10 anxiety items that overlapped with the state version of the State Trait Anxiety Inventory (Spielberger et al., 1970). See Table 1 and the [online supplemental materials](#) for more details.

⁷ Analytic strategies for analyzing such multilevel data have been given different names in a variety of fields and are known by a number of different terms such as the random effects or random coefficient models of economics, multilevel models of sociology and psychology, hierarchical linear models of education or more generally, mixed effects models (Fox, 2016).

Table 1

Calculating Multiple Measures of Internal Consistency Reliability Demonstrated on 10 Items From the Motivational State Questionnaire (msqR Data Set, $N = 3,032$)

Variable	10 anxiety items from the msqR data set										g	F1	F2	h2
	anxis	jtry	nervs	tense	upset	at.s-	calm-	cnfd-	cntn-	rlxd-				
anxious	1.00										.41	.60	.00	.53
jittery	.47	1.00									.47	.41	.00	.39
nervous	.54	.48	1.00								.43	.60	.00	.55
tense	.57	.48	.57	1.00							.54	.58	.00	.63
upset	.29	.16	.35	.45	1.00						.34	.32	.00	.22
at.ease-	.23	.24	.29	.35	.30	1.00					.66	.00	.47	.66
calm-	.28	.32	.31	.36	.23	.62	1.00				.69	.00	.31	.57
confident-	-.01	-.02	.08	.07	.17	.44	.31	1.00			.11	.00	.77	.61
content-	.07	.04	.13	.19	.29	.56	.45	.61	1.00		.31	.00	.74	.65
relaxed-	.27	.34	.30	.40	.29	.60	.56	.34	.45	1.00	.68	.00	.33	.57
SMC	.42	.37	.44	.52	.27	.55	.47	.40	.51	.47				
r_{ii}	.73	.69	.70	.77	.76	.63	.65	.70	.73	.60				

	Formula	Calculation	Reliability measure
Total variance = $V_X = \Sigma(R_{ij}) = 39.60$			
Total reliable item variance = $\Sigma r_{ii} = 6.97$	$g_{lb} = \frac{V_X - tr(R) + \Sigma(r_{ii})}{V_X}$	$\frac{39.60 - 10 + 6.97}{39.60}$	= .923
r best split (A = 1, 2, 5, 6, 8 vs B = 3, 4, 7, 9, 10) = .81	$\lambda_4 = \text{best split half} = \frac{2r_{ab}}{1 + r_{ab}}$	$\frac{2 \times .81}{1 + .81}$	= .895
Total common variance = $\Sigma h_i^2 = 5.37$	$\omega_i = \frac{V_X - tr(R) + \Sigma h_i^2}{V_X}$	$\frac{39.60 - 10 + 5.37}{39.60}$	= .883
Total squared multiple correlation $\Sigma(SMC) = 4.43$	$\lambda_6 = \frac{V_X - tr(R) + \Sigma(SMC)}{V_X}$	$\frac{39.60 - 10 + 4.43}{39.60}$	= .859
Average squared correlation = $\bar{r}^2 = \frac{\Sigma R_{ij}^2 - tr(R^2)}{n \times (n - 1)} = .137$	$\lambda_2 = \frac{V_X - tr(R) + \sqrt{\bar{r}^2 \times n/(n-1)}}{V_X}$	$\frac{39.6 - 10 + \sqrt{.137 \times 10/9}}{39.60}$	= .841
	$\alpha = \frac{n}{n-1} \frac{V_X - tr(R)}{V_X}$	$\frac{10}{9} \frac{39.60 - 10}{39.60}$	= .831
Average correlation = $\bar{r} = \frac{V_X - tr(V_X)}{n \times (n - 1)} = 0.329$	$\alpha = \frac{n\bar{r}}{1 + (n-1)\bar{r}}$	$\frac{10 \times .329}{1 + 9 \times .329}$	= .831
r worst split (A = 1-5 vs. B = 6-10) = .385	$\beta = \text{worst split half} = \frac{2r_{ab}}{1 + r_{ab}}$	$\frac{2 \times .385}{1 + .385}$	= .556
Sum of g loadings = 4.65 (bifactor)	$\omega_g = \frac{(\Sigma g_i)^2}{V_X}$	$\frac{4.65^2}{39.60}$	= .545
Sum of g loadings = 4.09 (Schmid-Leiman)	$\omega_h = \frac{(\Sigma g_i)^2}{V_X}$	$\frac{4.09^2}{39.60}$	= .422

Note. The row names for the items have been abbreviated for clarity. Negative signs following the item imply the item should be negatively scored. The 10 items may be thought of as measures of state anxiety. Five are positively scored, five negatively. General factor loadings (g) and group factor loadings were found from the `omegaSem` function, which applies a bifactor solution. The hierarchical solution from `omega` applies the Schmid-Leiman transformation and has slightly lower general factor loadings. Split half calculations were done by finding all possible splits of the test. Although the statistics shown are done by hand, they are all done automatically in various *psych* functions (see Table 2).

involved when answering trait like questions (Cattell, 1964; Chmielewski & Watson, 2009). State measures of affectivity probably involve episodic memory whereas trait measures of similar constructs (e.g., trait anxiety or neuroticism) likely tap semantic memory (Klein, Cosmides, Tooby, & Chance, 2002). With only two measures of state anxiety and one of trait anxiety, we cannot disentangle how much of the trait measure is state (Equation 9), but if we had more measures over longer periods of time we would be able to do so.

Alternate Forms

If we do not want to wait for a long time and we do not want to exactly repeat the same items, we can estimate reliability by

creating another test (an alternate form) that has conceptually similar but semantically different items. If measuring the same construct (e.g., arithmetic performance) we can subtly duplicate items on each form and even match for possible difficulty of order effects (a1: what is 6 + 3? and a2: what is 4 + 5? vs. b1: what is 3 + 6? and b2: what is 5 + 4?). Cattell (1964) discusses “Herringbone” consistency, which are essentially parallel forms: Each half of the test is made up of half of the items of multiple constructs, and each is duplicated in the other half (math, English, social studies). Although creating alternate forms by hand is tedious, it has become possible for ability items to generate alternate forms using computer Automatic Item Generation techniques (Embretson, 1999; Leon & Revelle, 1985; Loe & Rust, 2017).

Alternate forms given at the same time eliminate the effect of the specific item variance but do not remove any motivational state effect: Sometimes alternate forms can be developed when a longer test is split into multiple shorter tests. As an example of this, consider the *sai* data set discussed in the [online supplemental materials](#), which includes 20 items, 10 of which overlapped with the *msqR* data set and were used for our examples of test–retest and repeated measure reliability. The other 10 can be thought of as an alternate form of the anxiety measure and indeed correlate .74 with the target items from the *sai* and *msqR*. These correlations are less than when we actually repeat the same items by correlating the overlapping items of the *sai* and *msqR* (.85); are almost identical when we consider their short term dependability (.76); but less than estimates of internal consistency such as α or the average split half reliability (.83–.87).

Split Half (Adjusted for Test Length)

If we have gone to the trouble of developing two alternate forms for a concept and then administered both forms to a sample of participants, it is logical to ask what is the reliability of the composite formed from both of these tests. That is, if we have the correlation between two 5-item tests, what would be the reliability of the composite 10-item test? With a bit of algebra, we can predict it using a formula developed by [Spearman \(1910\)](#) and [Brown \(1910\)](#):

$$r_{xx} = \frac{2 \times r_{x_1x_2}}{1 + r_{x_1x_2}}. \quad (12)$$

It is important to note the correlation between the two parts ($r_{x_1x_2}$) is not the split half reliability, but is used to find the split half reliability (r_{xx}) found by the “Spearman-Brown prophecy formula” ([Equation 12](#))

Given that we have written n items and formed them into two splits of length $n/2$, what if we formed a different split? How should we split the items into two groups? Odd/even, first half/last half, randomly? This is a combinatorially difficult problem, in that there are $\frac{n!}{2(n/2)!(n/2)!}$ unique ways to split a test into two equal parts. While there are only 126 possible splits for the 10 anxiety items discussed above, this becomes 6,435 for a 16-item ability test, 1,352,078 for the 24-item EPI Extraversion Scale ([Eysenck & Eysenck, 1964](#)), and over 4.5 billion for a 36-item test. The split-Half function will try all possible splits for tests of 16 items or less, and then sample 10,000 splits for tests longer than that. The distribution of all possible splits for the 10 state anxiety items discussed earlier show that greatest split-half reliability is .92, the average is .87, and the lowest is .66 ([Figure 3](#) panel A). This is in contrast to all the possible splits of 16 ability items taken from the International Cognitive Ability Resource (ICAR; [Condon & Revelle, 2014](#)) collected as part of the SAPA project ([Revelle, Wilt, & Rosenthal, 2010](#); [Revelle et al., 2016](#)), where the greatest split-half reliability was .87, the average is .83, and the lowest is .73 ([Figure](#)

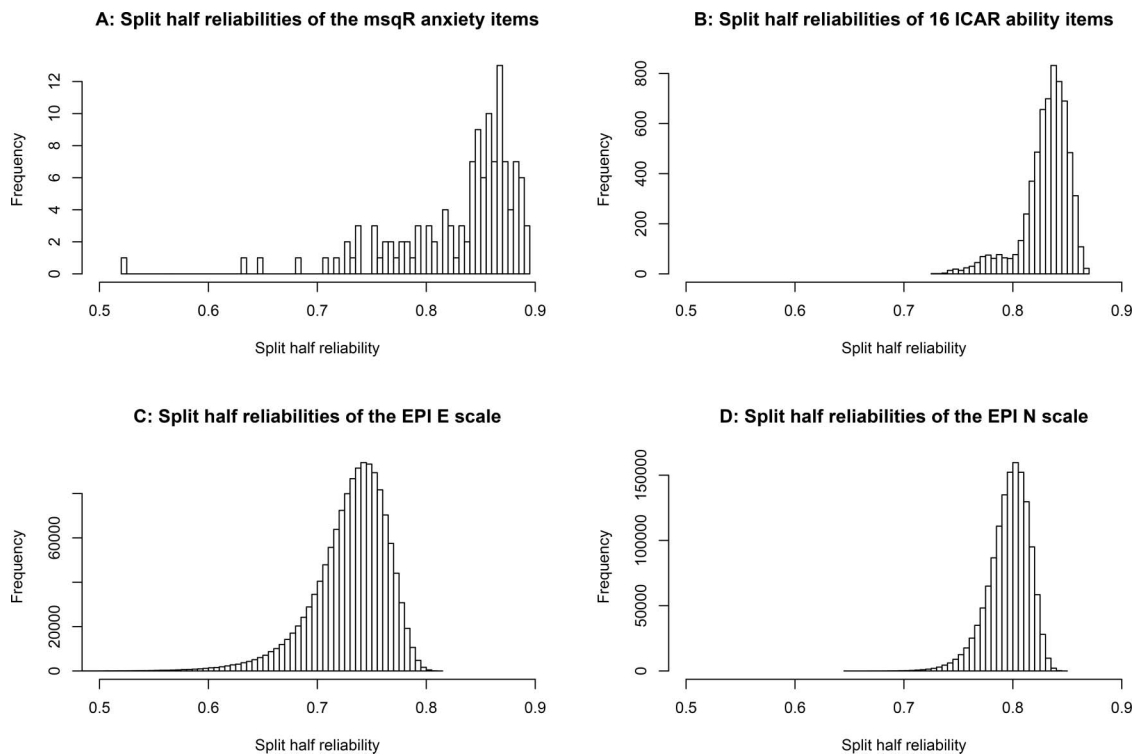


Figure 3. The distribution of 126 split-half reliabilities for the 10 state anxiety items (panel A) and the 1,352,078 splits of the 24 EPI Extraversion items (panel C) suggests that the tests are not univocal while that of the 6,435 splits of the ICAR ability items (panel B) and the 1,352,078 splits of the EPI N Scale (panel D) suggests greater homogeneity.

3 panel B). The 24 items of the EPI show strong evidence for nonhomogeneity, with a maximum split-half reliability of .81, an average of .73, and a minimum of .42 (Figure 3 part C). This supports the criticism that the EPI E Scale tends to measure the two barely related constructs of sociability and impulsivity (Rocklin & Revelle, 1981). The EPI N Scale, on the other hand, shows a maximum split-half of .85, a mean of .80, and a minimum of .65, providing strong evidence for a relatively homogeneous scale (Figure 3 part D). Examining these various splits is one way to understand the homogeneity of the test, for if the various splits differ a great deal (e.g., the EPI E Scale), this can be taken as a warning that the test is not unidimensional.

Internal Consistency and Domain Sampling

All of the above procedures are finding the correlation between two forms or occasions of a test. But what if there is just one form and one occasion? The approaches that consider just one test are collectively known as internal consistency procedures but also borrow from the concepts of domain sampling and can use the variance decomposition techniques discussed earlier. Some of these techniques (e.g., Cronbach, 1951; Guttman, 1945; Kuder & Richardson, 1937) were developed before advances in computational speed made it trivial to find the factor structure of tests, and were based upon test and item variances. These procedures (α , λ_3 , KR20) were essentially shortcuts for estimating reliability. The variance decomposition procedures continued this approach but expanded to be known as *generalizability theory* (Cronbach et al., 1963; Gleser et al., 1965; Vispoel, Morris, & Kilinc, 2018) and allow for the many reliability estimates discussed before.

There are a number of different approaches for estimating reliability when there is just one test and one time. The earliest was to split the test into two random split halves and then adjust the resulting correlation between these two splits using the Spearman-Brown prophecy formula (Brown, 1910; Spearman, 1910).

Unfortunately, as we showed in Figure 3, not all random splits produce equal estimates. If we consider all of the items in the test to be randomly sampled from some larger domain (e.g., trait-descriptive adjectives sampled from all words in the Oxford Unabridged Dictionary or sociability items sampled from a potentially infinite number of ways of being sociable), then we can think of the test as a sample of that domain. Because the item covariances should reflect just shared domain variance, but item variance will be an unknown mixture of domain and specific and error variance, the amount of domain variance in a test would vary as the square of the number of items in the test times the average covariance of the items in the test. Considering items as measuring ability, Kuder and Richardson (1937) proposed several estimates of the reliability of the average split half, with their most well-known being their 20th equation (and thus known as KR20).

A more general form of KR20 allows items to be not just right or wrong and thus corrects for the sum of the individual item variances. This is known as coefficient α (Cronbach, 1951) as well as λ_3 (Guttman, 1945)

$$\alpha = \lambda_3 = \frac{n}{n-1} \frac{V_t - \sum v_i}{V_t} \quad (13)$$

where V_t is the total test variance, v_i is the variance for a particular item, and there are n items in the test.

Alpha and λ_3 may be thought of as the correlation of a test with a nonexistent test just like it. That is, they are estimates of reliability based upon a fictitious parallel test. Alpha estimates the correlation between the observed test and its hypothetical twin by assuming that the average covariance within the observed test is the same as the average covariance between items of the observed and with the hypothetical test. It is correct to the extent that the average interitem correlation correctly estimates the amount of domain score variance (an unknown mixture of trait and state variance) in each item. But this is only correct if all the items have equal covariances and differ only in their observed variances. In this case they are said to be τ equivalent (Lord & Novick, 1968), which is a fancy way of saying that they all have equal covariances with the latent score represented by the test and have equal factor loadings on the single factor of the test. This is very unlikely in practice and deviations from this assumption will lead to α underestimating reliability (Teo & Fan, 2013).

In addition to λ_3 , Guttman (1945) considered five alternative ways of estimating reliability by correcting for the error variance of each item. All of these equations recognize that some of the item is reliable variance, the problem is how much? λ_3 and α assume that the average item covariance is a good estimate, λ_6 uses the Squared Multiple Correlation (smc) for each item as an estimate of its reliable variance, while λ_2 uses a function of the average squared item covariance. λ_4 is just the maximum split-half reliability.

One advantage of using the mean item covariance is that it can be identified from an analysis of variance perspective rather than actually finding all the intercovariances. That is, just decompose the total test variance into three components: the between person variance σ_p^2 , the between item variance, σ_i^2 , and the interaction of Person $\times$ Item, σ_e^2 . Then reliability is just $1 - \frac{\sigma_e^2}{\sigma_p^2 + \sigma_i^2 + \sigma_e^2}$ (Feldt, Woodruff, & Salih, 1987; Hoyt, 1941). By expressing it in this manner, Feldt et al. (1987) were able to derive an F distribution for α , and thus a means for finding confidence intervals. This is implemented as the `alpha.ci` function in the *psych* package. Alternative procedures for the confidence interval for α have been developed by Duhachek and Iacobucci (2004). Perhaps the biggest advantage to the variance approach to KR20, α , or λ_3 was that in the 1930s–1950s calculations were done with desk calculators rather than computers and it was far simpler to find the n item variances and one total test variance than it was to find the $n(n-1)/2$ item covariances. In the modern era, such shortcuts are no longer necessary.

Two Problems With α

Although easy to calculate from just the item statistics and the total score, α and λ_3 are routinely criticized as poor estimates of reliability because they do not reflect the structure of the test (Bentler, 2009; Cronbach & Shavelson, 2004; Revelle & Zinbarg, 2009; S. Green & Yang, 2009; Sijtsma, 2009). Perhaps because the ability to find α is available in easy-to-use software packages, it is routinely used. This is unfortunate; except for very rare conditions, α is both an underestimate of the reliability of a test (because of the lack of τ equivalency, Bentler, 2009, 2017; Sijtsma, 2009) and an overestimate of the fraction of test variance that is associated with the general variance in the test (Revelle, 1979; Revelle & Zinbarg, 2009; Zinbarg, Revelle, Yovel, & Li, 2005). As we show in the

online supplemental materials (see Table S2), α provides no information about the constancy or stability of the test. For our mood items, α (.83–.87) exceeded the short term constancy estimates (.42–.76) and greatly exceeded the 2-day stability coefficients (.36–.39). For the trait measures (particularly of impulsivity), the low α (.51) did not reflect the relatively high (.70) 2–4 week stability of the measures. That is to say, knowing α told us nothing about test–retest constancy or stability.

If not an estimate of reliability, does α measure internal consistency? No. For it is just a function of the number of items and the average correlation between the items. It is not a function of the unidimensionality of the test. It is easy to construct example tests with equal α values that reflect one test with homogenous items, two slightly related subtests, or even two unrelated subtests each with homogeneous items (see, e.g., Revelle, 1979; Revelle & Wilt, 2013).

Model-Based Estimates

That “internal consistency” estimates do not reflect the internal structure of the test becomes apparent when we apply “model based” techniques to examine the factor structure of the test. These procedures actually examine the correlations or covariances of the items in the test. Thanks to improvements in computational power, the task of finding correlations and the factor structure of a 10-item test has been transformed over the past two generations from being a summer research project for an advanced graduate student to an afternoon homework assignment for undergraduates. Using the latent variable modeling approach of factor analysis, these procedures decompose the test variance into that which is common to all items ($\mathbf{g}$, a general factor), that which is specific to some items (orthogonal group factors, $\mathbf{f}$) and that which is unique to each item (typically confounding specific, $\mathbf{s}$, and error variance, $\mathbf{e}$). Many researchers have discussed this approach in great detail (e.g., Bentler, 2017; McDonald, 1999; Revelle & Zinbarg, 2009; Zinbarg et al., 2005) and we just summarize the main points here. Most importantly for applied researchers, as we show in the online supplemental materials, model-based techniques are just as easy to implement in modern software as are the more conventional approaches.

The observed score on a test item may be modeled in terms of the sum of the products of factor scores ($\mathbf{g}$, $\mathbf{f}$, $\mathbf{s}$, $\mathbf{e}$) and loadings ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{D}$) on these factors:

$$\mathbf{x} = \mathbf{c}\mathbf{g} + \mathbf{A}\mathbf{f} + \mathbf{D}\mathbf{s} + \mathbf{e} \quad (14)$$

Because the reliable variance of the test is that which is not error, the reliability of a test with standardized items should be

$$\omega_t = \frac{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1} + \mathbf{1}'\mathbf{A}\mathbf{A}'\mathbf{1}}{V_x} = 1 - \frac{\Sigma(1 - h_i^2)}{V_x} = 1 - \frac{\Sigma u_i^2}{V_x} \quad (15)$$

where h_i^2 is the item communality and u_i^2 is the item uniqueness. The percentage of the total variance that is due to the general factor (ω_g , McDonald, 1999) is

$$\begin{aligned} \omega_g &= \frac{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1}}{V_x} = \frac{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1}}{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1} + \mathbf{1}'\mathbf{A}\mathbf{A}'\mathbf{1} + \mathbf{1}'\mathbf{D}\mathbf{D}'\mathbf{1} + \mathbf{1}'\mathbf{e}\mathbf{e}'\mathbf{1}} \\ &= 1 - \frac{(\Sigma c_i)^2}{V_x}, \end{aligned} \quad (16)$$

where the total test variance (V_x) is the sum of the elements of all the item variances and covariances and $(\Sigma c_i)^2$ is the squared sum of the loadings on the general factor.

Normally, the specific item variance is confounded with the residual item (error) variance, but if we have a way of estimating the specific variance by examining the correlations with items not in the test, (e.g., repeated items, Wood et al., 2017) then we can include it as part of the reliable variance (Bentler, 2017):

$$\begin{aligned} \omega_t &= \frac{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1} + \mathbf{1}'\mathbf{A}\mathbf{A}'\mathbf{1} + \mathbf{1}'\mathbf{D}\mathbf{D}'\mathbf{1}}{V_x} \\ &= \frac{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1} + \mathbf{1}'\mathbf{A}\mathbf{A}'\mathbf{1} + \mathbf{1}'\mathbf{D}\mathbf{D}'\mathbf{1}}{\mathbf{1}'\mathbf{c}\mathbf{c}'\mathbf{1} + \mathbf{1}'\mathbf{A}\mathbf{A}'\mathbf{1} + \mathbf{1}'\mathbf{D}\mathbf{D}'\mathbf{1} + \mathbf{1}'\mathbf{e}\mathbf{e}'\mathbf{1}}. \end{aligned} \quad (17)$$

Unfortunately, in his development of ω , McDonald (1999) refers to two formulae (6.20a and 6.20b) one for ω_t and one for ω_g , and calls them both ω (Zinbarg et al., 2005). These two coefficients are very different, for one is an estimate of the total reliability of the test (ω_t), the second is an estimate of the amount of variance in the test due to single, general factor (ω_g). Then to make it even more complicated, there are two ways to find the general factor. One method uses a bifactor solution (Holzinger & Swinford, 1937; Reise, 2012; Rodriguez, Reise, & Haviland, 2016) using structural equation modeling software (e.g., lavaan, Rosseel, 2012); the other extracts a higher order factor from the correlation matrix of lower level factors and then applies a transformation developed by Schmid and Leiman (1957) to find the general loadings on the original items. The bifactor solution (ω_g) tends to produce slightly larger estimates than the Schmid-Leiman procedure (ω_h) because it forces all the cross loadings of the lower level factors to be 0. Following Zinbarg et al. (2005), we designate the Schmid-Leiman solution as ω_h recognizing the hierarchical nature of the solution. Both approaches are implemented in the *psych* package.

An important question when examining a hierarchical structure is how many group factors to specify when calculating ω_h ? The Schmid-Leiman procedure is defined if there are three or more group factors, and with only two group factors the default is to assume that they are both equally important (Zinbarg, Revelle, & Yovel, 2007). While the Schmid-Leiman approach is exploratory, the bifactor approach is a confirmatory model that requires specifying which variables load on each group factor.

How do these various approaches differ and what difference does it make? If we want to correct observed correlations for attenuation by using Equation 3, then underestimating reliability will lead to serious overestimation of the true validity of a measure. This is why there has been so much work on trying to estimate the greatest lower bound of reliability (e.g., Bentler, 2017). In this case if α underestimates reliability, it is a poor measure to use when correcting for attenuation. In addition, many of the conventional measures do not reflect the percentage of total variance that is actually common to all of the items in the test. For factor analytic approaches, this is only done by ω_g and ω_h ; for non-model-based procedures this is the worst split-half reliability (β).

In order to show how these various approaches can give very different values, we consider a real-life data set consisting of the 10 anxiety items discussed in the online supplemental materials. We show the correlation matrix as well as different reliability estimates in Table S5 in the online supplemental materials. Even

though the greatest reliability estimates exceed .90, it is important to remember that this does not imply anything about the stability of the measure, which is just .30 after 2 days. (See Table S1 in the online supplemental materials.)

The ω , based value of .88 agrees closely with the greatest split-half of .89 or the duplicate item estimate of .92. These are all estimates of the total reliable variance. The worst split-half .56 and ω_g values of .55 suggest that slightly less than 60% of the test reflects one general factor of anxiety. The difference between the .9 and the .6 values suggest that roughly 30% of the total test variance is due to the positively worded versus negatively worded group variance. That is, roughly 2/3 of the reliable test variance represents one construct, and about 1/3 represents something not shared with the total test. Note that the α of .83 does not provide as much information.

Tetrachoric, Polychoric, and Pearson Correlations

Test scores are typically the sum or average of a set of individual items. Each item is thought to reflect some underlying latent trait. Because the items are not continuous but rather are dichotomous or polytomous, the normal Pearson interitem correlation will be attenuated from what would be observed if it were possible to correlate the latent scores associated with each item. The latent correlation can be estimated using tetrachoric or polychoric correlations, which find what a (latent) continuous bivariate normal correlation would be given the observed pairwise cell frequencies. The use of such correlations is recommended when examining the structure of a set of items using factor analysis, for a clearer structure will appear and artificial difficulty factors will not be found. However, the temptation to use tetrachoric or polychoric correlations when finding the reliability of a test using any of the formulas in Table S5 in the online supplemental materials (e.g., McNeish, 2018) should be resisted, for this will lead to overestimates of the amount of variance in the observed test made up of the observed items (Revelle & Condon, 2018).

Reliability and Test Length

With the exception of the worst split-half reliability (β) and hierarchical ω (estimated either by a bifactor approach, ω_g or the Schmid-Leiman procedure ω_h) all of the reliability estimates in online Table S5 are functions of test length and will tend asymptotically toward 1 as the number of items increases. Examining the equations in online Table S5 makes this clear: each method replaces the diagonal of the test, $tr(V_x)$, with the sum of some estimate based on the item reliability (r_{ii} , h^2 , the SMC, $\sqrt{r_{ii}}$, or $\bar{r}_{ij}$) and then compares this adjusted test variance to the total test variance. But as the number of items in the test increases, the effect of the diagonal elements becomes less as a fraction of the total test variance. Thus, the limit of the glb, λ_4 , ω , λ_6 , λ_2 , α as n increases to infinity is 1, ω_h does not have this problem as it will increase toward the limit of $\omega_{g\infty} = \frac{1cc'1}{V_x} = \frac{1cc'1}{1cc'1 + 1AA'1 + 1DD'1}$. When comparing reliabilities between tests of different lengths, it is useful to include the reliability of each test as if they were just one item each. In the case of α , $\alpha_1 = \bar{r}_{ij}$. Other single item reliability measures are the average item test retest ($glb_1 = \bar{r}_{ii}$), the average communality ($\omega_{t_1} = \bar{h}^2$), the average SMC ($\lambda_{6_1} = SMC_i$), or the square root of the average squared correlation ($\lambda_{2_1} = \sqrt{\bar{r}_{ii}}$).

Generalizability Theory

Most discussions of reliability consider reliability as the correlation of a test with a test just like it. Test-retest and alternate form reliabilities are the most obvious examples. Internal consistency measures are functionally estimating the correlation of a test with an imaginary test just like it. These estimates are based upon the patterns of correlations of the items within the test. An alternative approach makes use of Analysis of Variance procedures to decompose the total test variance into that due to individuals, to items, to time, relevant interactions, and to residual (Cronbach et al., 1963; Gleser et al., 1965; Shavelson, Webb, & Rowley, 1989; Vispoel et al., 2018). We have already discussed this in the context of test-retest reliability. This technique is most frequently applied to the question of the reliability of judges who are making ratings of targets, but the logic can be applied equally easily to item analysis.

Reliability of Raters

Consider the case where we are rating numerous subjects with only a few judges. We might do a small study first to determine how much our judges agree with each other, and depending upon this result, decide upon how many judges to use going forward. As an example, examine the data from five judges (raters) who are rating the anxiety of 10 subjects (Table S5 in the online supplemental materials). If raters are expensive, we might want to use the ratings of just one judge rather than all five. In this case, we will want to know how ratings of any single judge will agree with those from the other judges. In this case, differences in leniency (the judges' means) between judges will make a difference in their judgments. In addition, different judges might use the scale differently, with some having more variance than others. We also need to think about how we will use the judges. Will we use their ratings as given, will we use their ratings as deviations from their mean, or will we pool the judges? All of these choices lead to different estimates of generalizability. Shrout and Fleiss (1979) provide a very clear exposition of three different cases and the resulting equations for reliability (see Equations 17–19 in the online supplemental materials). Although they express their treatment in terms of Mean Squares derived from an analysis of variance (e.g., the `aov` function in R), it is equally easy to do this with variance components estimated using a mixed effects linear model (e.g., `lmer` from the `lme4` package [Bates et al., 2015] in R). Both of these procedures are implemented in the `ICC` function in the `psych` package. This is discussed in more detail in the online supplemental materials.

The intraclass correlation is appropriate when ratings are numerical, but sometimes ratings are categorical (particularly in clinical diagnosis or in evaluating themes in stories). This then leads to measures of agreement of nominal ratings. Rediscovered multiple times and given different names (Conger, 1980; Scott, 1955; Hubert, 1977; Zapf, Castell, Morawietz, & Karch, 2016), perhaps the most standard coefficient is known as Cohen's Kappa (Cohen, 1960, 1968), which adjusts observed proportions of agreement by the expected proportion:

$$\kappa = \frac{p_o - p_e}{1 - p_e} = \frac{f_o - f_e}{N - f_e} \quad (18)$$

where $p_o = \frac{f_o}{N}$ is the observed proportion (p_o) or frequency of agreement (f_o) between two observers, and $p_e = \frac{f_e}{N}$ is the expected

proportion or frequency of agreement (f_e ; Cohen, 1960; see Table S7 in the online supplemental materials). Because raw agreements will reflect the base rates of judgments, κ corrects for the expected agreement on the assumption of independence of the raters. Thus, if two raters each use one category 60% of the time, we would expect them to agree by chance 36% of the time in their positive judgments and 16% in their negative judgments. Various estimates of correlations of nominal data have been proposed and differ primarily in the treatment of the correction for chance agreement (Feng, 2015). Thus, κ adjusts for differences in the marginal likelihood of judges, while Krippendorff's α_k does not (Krippendorff, 1970, 2004). To Krippendorff (2004) this is a strength of α_k , but to Fleiss it is not (Krippendorff & Fleiss, 1978).

If some disagreements are more important than others, we have weighted κ , which with appropriate weights is equal to the intraclass correlation between the raters (Cohen, 1968; Fleiss & Cohen, 1973). For multiple raters, the average κ is known as Light's κ (Conger, 1980; Light, 1971).

Real life examples of a range of κ values are given by Freedman et al. (2013) in a discussion of the revised *Diagnostic and Statistical Manual of Mental Disorders*, where the κ values for clinical diagnoses range from *very good agreement* ($>.60$) for major neurocognitive disorders or posttraumatic stress disorder, to *good* (.40–.60) for bipolar II, or schizophrenia, to *questionable agreement* (.2–.4) for generalized anxiety or obsessive-compulsive disorder, to values that did not exceed the confidence values of 0. When comparing the presence or absence of each of five narrative themes in a life story interview, Guo, Klevan, and McAdams (2016) report how two independent raters of each of 12 different interview segments showed high reliability of judgments with κ values ranging from .61 (did the story report early advantage) to .83 (did the story discuss prosocial goals).

Multilevel Reliability

With the introduction of cell phones and various apps, it has become much easier to collect data within subjects over multiple occasions (e.g., A. S. Green, Rafaeli, Bolger, Shrout, & Reis, 2006; Bolger & Laurenceau, 2013; Mehl & Conner, 2012; Wilt et al., 2011, 2017). This has taken us from the daily diary to multiple mood measures taken multiple times per day. These techniques lead to fascinating data, in that we can examine patterns of stability and change within individuals over time. These intensive longitudinal methods (Walls & Schafer, 2006) “captures life as it is lived” (Bolger, Davis, & Rafaeli, 2003). They also lead to important questions about reliability. How consistent is one person over time? How stable are the differences between people over time? The same decomposition of variance techniques discussed for raters and for generalizability theory can be applied to an analysis of temporal patterns of reliability (Revelle & Wilt, 2019; Shrout & Lane, 2012). That is to say, we decompose the responses into variance components due to stable individual differences (σ_p^2), to differences due to time (σ_t^2), to the interaction of person by time effects ($\sigma_{p \times t}^2$), and to residual error σ_e^2 . Shrout and Lane (2012) give the SPSS and SAS syntax to do these calculations. In R this merely requires calling the `multilevel.reliability` function in *psych*. In the interest of space, we refer the interested reader to Shrout and Lane (2012); Revelle and Wilt (2019); and the examples in the online supplemental materials.

When doing multilevel reliability, it is straightforward to find the reliability of each individual subject over items and over time. People are not the same and the overall indices do not reflect how some subjects show a very different pattern of response. The `multilevel.reliability` function returns reliability estimates for each subject over time, as well as the six estimates shown in the online supplemental materials for 77 subjects on our 10 anxiety items across four time points.

Composite Scores

The typical use of reliability coefficients is to estimate the reliability of relatively homogeneous tests. Indeed, the distinctions made between ω_h , α , and ω_t are minimized if the test is completely homogeneous. But if the test is intentionally made up of unrelated or partly unrelated content, then we need to consider the reliability of a composite score. A composite is sometimes referred to as a stratified test, where the strata may be difficulty or content based (Cronbach, Schönemann, & McKie, 1965). The stratified reliability (ρ_{xx_s}) of a composite test is found by replacing the variance of each subtest in the total test with its reliable variance and then dividing the resulting sum by the total test variance:

$$\rho_{xx_s} = \frac{V_t - \sum v_i + \sum p_{xx_i} v_i}{V_t} \quad (19)$$

where ρ_{xx_i} is reliability of the subtest and v_i is the variance of the subtest (Rae, 2007). Conceptually, this approach is very similar to ω_t (McDonald, 1999).

A procedure for weighting the elements of the composite to maximize the reliability of composite scores is discussed by Cliff and Caruso (1998), who suggest this as a procedure for Reliable Components Analysis (RCA) which they see as an alternative to a EFA or PCA.

Reliability of a Difference Score

Logically similar to the reliability of a composite is the reliability of a difference score (Equation 20). Sometimes researchers want to find the difference between two scores (e.g., verbal and spatial ability or anxiety and depression). Even though the two tests themselves are highly reliable (ρ_{xx} , ρ_{yy}), if they also have a high correlation, (r_{xy}) the reliability of the difference will be substantially lower. Indeed, if the correlation between the two scales matches their reliability, the reliability of the difference will be 0. Given this reduction in reliability, individual differences in change or in pattern should be interpreted cautiously. We give an example of this problem when comparing the difference of two cognitive tests (i.e., verbal vs. spatial reasoning) in the online supplemental materials.

$$\rho_{x-y} = \frac{\rho_{xx} + \rho_{yy} - 2r_{xy}}{2 \times (1 - r_{xy})} \quad (20)$$

Beyond Classical Test Theory

Reliability is a joint property of the test and the people being measured by the test (refer back to Equation 2). For fixed amount of error, reliability is a function of the variance of the people being assessed. Scores from a test of ability will be reliable if given to a random sample of 18–20 year olds, but much less reliable if given

to students at a particularly selective college because there will be less between-person variance. The reliability of scores of emotional stability will be higher if given to a mixture of psychiatric patients and their spouses than it will be if given just to the patients. That is, reliability is not a property of test scores independent of the people taking it. This is the basic concept of Item Response Theory (IRT), called by some the “new psychometrics” (Embretson, 1996, 1999; Embretson & Reise, 2000), and which models the individual’s patterns of response as a function of parameters (discrimination, difficulty) of the item.

By focusing on item difficulty (endorsement frequency) it is possible to consider the range of application of our scores. Items are most informative if they are equally likely to be passed or failed (endorsed or not endorsed). But this can only be the case for a particular person taking the test and cannot be the case for a person with a higher or lower latent score. Although test scores are maximally reliable if all of the items are equally difficult, such scores will not be very discriminating at any other than at that level (Loevinger, 1954). Thus, we need to focus on spreading out the items across the range to be measured.

The essential assumptions of IRT is that items can differ in how hard they are, as well as how well they measure the latent trait. Although seemingly quite different from classical approaches, there is a one-to-one mapping between the difficulty and discrimination parameters of IRT and the factor loadings and item response thresholds found by factor analysis of the polychoric correlations of the items (Kamata & Bauer, 2008; Markon, 2013; McDonald, 1999). The relationship of the IRT approach to classical reliability theory is given a very clear explication by Markon (2013), who examines how test information (and thus the reliability) varies by subject variance as well as trait level. A test can be developed to be reliable for certain discriminations (e.g., between psychiatric patients) and less reliable for discriminating between members of a control group. The particular strength of IRT approaches is the use in tailored or adaptive testing where the focus is on the reliability for a particular person at a particular level of the latent trait. (See the discussion of IRT in the [online supplemental materials](#), particularly [Figure S3](#), which shows how reliability differs as a function of latent score).

The Several Uses of Reliability

Reliability is measured for at least three different purposes: correcting for attenuation, estimating expected scores, and providing confidence intervals around these estimates. When comparing test reliabilities, it is useful to remember that reliability has non-linear relations with the standard error as well as with the signal/noise ratio (Cronbach et al., 1965). That is, seemingly small differences in reliability between tests can reflect large differences in the ratio of reliable signal to unreliable noise or the size of the standard error of measurement. Consider the signal-to-noise ratio of tests with reliability of .7, .8, .9, and .95.

$$\frac{\text{Signal}}{\text{Noise}} = \frac{\rho_{xx}}{1 - \rho_{xx}}.$$

Thus an improvement in reliability from $.7(\frac{7}{3} = 2.33)$ to $.8(\frac{8}{2} = 4)$ is a much smaller change in signal to noise than that from $.8$ to $.9(\frac{9}{1} = 9)$ which in turn is much less than from $.9$ to $.95(\frac{95}{5} = 19)$.

Corrections for Attenuation

Reliability theory was originally developed to adjust observed correlations between related constructs for the error of the measurement in each construct (Spearman, 1904b). Such corrections for attenuation were perhaps the primary purpose behind reliability and are the reason that some recommend routinely correcting for reliability when doing meta-analyses (Schmidt & Hunter, 1999). However such a correction is appropriate only if the measure is seen as the expected value of a single underlying construct. Examples of when the expected score of a test is not the same as the theoretical construct that accounts for the correlations between the observed variables include chicken sexing (Lord & Novick, 1968) or the diagnosis of Alzheimers (Borsboom & Mellenbergh, 2002). Modern software for Structural Equation Modeling (e.g., Rosseel, 2012) models the pattern of observed correlations in terms of a measurement (reliability) model as well as a structural (validity) model.

Reversion to Mediocrity

Given a particular observed score, what do we expect that score to be if the measure is given again? That high scores decrease and low scores increase is just a function of the reliability of the test (Equation 5) with larger drops and gains for extreme scores than for moderate scores. Although expected, these regression effects can mislead those who do not understand reliability and lead to surprise when successful baseball players are less successful the next year (Schall & Smith, 2000) or when poorly performing pilots improve but better performing pilots get worse (Kahneman & Tversky, 1973). That superior performance is partly due to good luck is hard for high performers to accept and that poor performance is partly due to bad luck leads to false beliefs about the lack of effect for rewards and the strong effect of punishment (Kahneman & Tversky, 1973).

Confidence Intervals, Expected Scores, and the Standard Error

Not only does reliability affect the regression toward the mean, it also affects the precision of measurement. The standard error of measurement is a function of sample variability as well as the reliability (Equation 4). Confidence intervals for expected scores are symmetric around the expected score (Equation 5), and therefore are not symmetric around the observed score. Combining these two equations and taking into account the variance of the expected score we see that the confidence interval for an expected score based upon an observed score, X , with a sample variance of V_x , mean of $\bar{X}$ (and thus deviation score, x and estimated reliability of ρ_{xx} is

$$\rho_{xx}(x) - \sqrt{\rho_{xx}V_x(1 - \rho_{xx})} < \rho_{xx}(x) < \rho_{xx}(x) + \sqrt{\rho_{xx}V_x(1 - \rho_{xx})}. \quad (21)$$

Estimating and Reporting Reliability

We have included many equations and referred to many separate R functions. What follows is a brief summary with an accompanying flowchart that we presented earlier (see [Table 2](#)).

Table 2
Steps Toward Reliability Analysis: Choosing the Appropriate R Function to Find Reliability

Steps	Statistic	R function
Preliminaries		
Hypothesis development		
Data collection		
Data input		<code>read.file</code>
Data screening		
Descriptive statistics	$\mu, \sigma, \text{range}$	<code>describe</code>
Analysis of internal structure		
Exploratory factor analysis	$\mathbf{R} = \mathbf{F}\Phi\mathbf{F}' + \mathbf{U}^2$	<code>fa</code>
Hierarchical structure	ω_h, ω_t	<code>omega, omegaSem</code>
Confirmatory factor analysis		<code>lavaan::cfa</code>
Estimation of various reliabilities		
Items (dichotomous, polytomous or continuous)		
One occasion		
General factor saturation	ω_h	<code>omega</code>
Total common variance	ω_t	<code>omega</code>
Average interitem r	$\bar{r}_{ij}$	<code>omega, alpha</code>
Median interitem r		<code>omega, alpha</code>
Mean test retest (tau equivalent)	α, λ_3	<code>omega, alpha</code>
Smallest split half reliability	β	<code>splitHalf, iclust</code>
Greatest split half reliability	λ_4	<code>splitHalf, guttman</code>
Two occasions		
Test–retest correlation	r	<code>cor</code>
Variance components	$\sigma_p^2, \sigma_i^2, \sigma_e^2$	<code>testRetest</code>
Multiple occasions		
Within subject reliability	α	<code>multilevel.reliability</code>
Variance components	$\sigma_p^2, \sigma_i^2, \sigma_e^2$	<code>multilevel.reliability</code>
Ratings (Ordinal or interval)		
Single rater reliability	$ICC_{1..31}$	<code>ICC</code>
Multiple rater reliability	$ICC_{1..3k}$	<code>ICC</code>
Ratings (Categorical)		
Two raters	κ	<code>cohen.kappa</code>

Note. All functions except for the `cfa` and `cor` function are in the *psych* package.

Preliminary Steps

The most important question to ask should be done before collecting the data: what are we trying to measure and how are we trying to measure it? Does the measure to be analyzed represent a single construct or is the factor structure more complicated? The next question is: who are the subjects of interest? The reliability of test scores is not a property of the test, but a joint function of the people taking the test and of the test itself. Thus specifying the latent construct and the population of interest is essential before collecting and analyzing data.

Once one has decided what to measure, the test items must be given to willing (and one can hope interested) participants. Steps should be taken to ensure participant involvement. Measures to take include the classic issues of data screening. Subjects who respond too rapidly or carelessly will not provide reliable information (Wood et al., 2017). If response times are available, it is possible to screen for implausibly fast responses. If items are repeated in the same session, it is also possible to screen for temporal consistency (DeSimone, 2015; Wood et al., 2017).

Type of Measurement and Tests for Unidimensionality

Is the test given more than once? Is it given many times? Are the data based upon item responses or ratings? Are the data categorical, dichotomous, polytomous, or continuous? For the latter three,

examining the structure of the correlations should be done to confirm the factor structure is as expected. A unidimensional scale would be expected to have just one large factor. More typical scales will probably have some subgroups that can be explored using hierarchical or bifactor models. If the data are dichotomous or polytomous, using the latent variable correlations estimated by tetrachoric or polychoric correlations will show the structure more clearly. However, when estimating the reliability of the resulting scores, the statistics should be based upon the Pearson correlations.

Which Reliability to Estimate

As we have discussed before, there is no one reliability estimate. If giving just one test on one occasion we need to rely on internal consistency measures: ω_h , β and the worst split-half reliability are estimates of the amount of general factor variance in the test scores. Simulations suggest that for very low levels of general factor saturation that the EFA based ω_h is positively biased and that a CFA based estimate (ω_g) is more accurate. ω_t is a model based estimate of the Greatest Lower Bound of the total reliability of a test as is the best split-half reliability (λ_4). If the items are repeated within one form, the *glb* can be found based upon the item test–retest values.

If tests are given twice, then test–retest measures *dependability* over the short term or *stability* over a longer term. Variance decomposition techniques can be used to estimate how much

variance is due to individuals, to the items, and to changes over time.

If tests are given many times, then multiple measures of reliability are relevant, each implying a different generalization: is time treated as fixed or random effect, are items seen as fixed or random. A powerful addition to this design is that reliability over time can be found for each subject as well as all of the subjects. The scores of some subjects may be much more reliable than others.

If the measures are not items, but rather raters, and we want to know the limits of generalizability of the raters to different raters, or for pooled raters, we can find estimates of the intraclass correlations. There are several of these; all can be estimated the same way.

These many forms of reliability coefficients (see Table 2) may all be found in the *psych* package (Revelle, 2019a) for the open source statistics environment, R (R Core Team, 2019). *psych* was specially developed for personality-oriented psychologists to be both thorough and easy to use. Although some of these statistics are available in commercial software packages, the *psych* package provides them all in one integrated set of functions. See Rstudio (RStudio Team, 2016) for a convenient interface. We show the specific commands to use to find all of these coefficients in the online supplement materials to this article.

Conclusions

Although we have used many equations to discuss it and many ways to estimate it, at its essence, reliability is a very simple concept: Reliability is a property of the test scores and is the correlation of a test with a test just like it, or alternatively, the fraction of the test score variance which is not due to error. Unfortunately, there is not just one reliability that needs to be reported, but rather a variety of coefficients, each of which is most appropriate for certain purposes. Are we trying to generalize over items, over time, over raters? Are we estimating unidimensionality, general factor saturation, or total reliable variance? Each of these questions leads to a different estimate (see Table 2). So rather than ask, What is the reliability?, we should ask, Which reliability and reliability for what?

The initial appeal of α or KR20 reliability estimates were that they were simple to calculate in the precomputer era. But this has not been the case for the past 60 years. The continued overuse of α is probably due to the ease of calculation in common commercial software. But with modern, open source software such as R, this is no longer necessary. Alpha, ω_h , ω_r , minimum (B) and maximum (λ_4) split halves, six ICCs, and six repeated measure reliabilities are all available with one or two simple commands. (See the online supplement materials for a guided tour.) It should no longer be acceptable to report one coefficient that is only correct if all items are exactly equally good measures of a construct. Readers are encouraged to report at least two coefficients (e.g., ω_h and ω_r) and then discuss why each is appropriate for the inference that is being made. They are discouraged from reporting just α unless they can justify the assumptions implicit in using it (i.e., τ equivalence and unidimensionality). When reporting the reliability of raters, it is useful to report all six ICCs and then explain why one is most appropriate. Similarly, when reporting multilevel reliabilities, an

awareness of what generalizations one wants to make is required before choosing between the six possible indices.

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